ME 249 Project report

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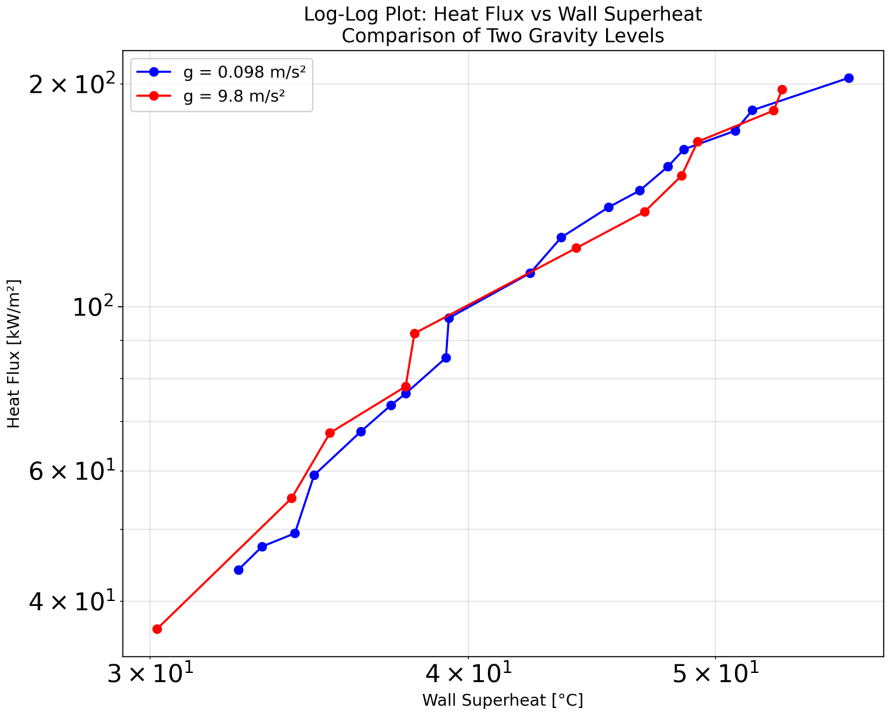
October 6, 2025

Task 1.1.a

Compared to the array in the Appendix on the last page of the instruction, the first value was not appended correctly. The provided code contained a value of [44.2] where the provided Appendix held a value of [44.1]. This was appended correctly to match the Appendix array.

Task 1.1.b

The log-log plot was generated using both numpy and matplotlib to append the ydata from part 1.1.a. Both g=0.098 and 0.8 m/s^2 were used to mask the gravity portion of the data to generate the plot.



Task 1.2

This task involved implementing a genetic algorithm to fit experimental heat-transfer data. To evaluate the algorithm's performance, four distinct sets of correlation constants were defined and compared. The provided source code was modified to include the parameter groups summarized in *Table 1*, corresponding to different combinations of n-values.

	n1	n2	n3
Group 1	0.00027	4.0	0.063
Group 2	0.0005	3.5	0.08
Group 3	0.001	3.0	0.1
Group 4	0.0005	3.3	0.056

Table 1. Correlation constants for genetic-algorithm comparison between four data groups.

Computation of RMS and Percent Error

In this routine, the fractional error was computed for each data point, and the RMS error was obtained as the square root of the mean of squared fractional errors. The RMS error was then expressed as a percentage to facilitate comparison with mean and maximum percent errors reported in the results.

$$RMS\ Error\ =\ sqrt(\ (1\ /\ N_D)\ *\ \Sigma\ [\ (q_data, i\ -\ q_pred, i)\ /\ q_data, i\ ]^2\ ) \qquad (1)$$

The following plots illustrate the convergence behavior of the genetic algorithm and the relationship between measured and predicted heat flux for each group. The left plot shows the evolution of the constants (n_1, n_2, n_3) and mean error with generation number, while the right plot presents a log–log comparison between predicted and measured heat fluxes

Group 1

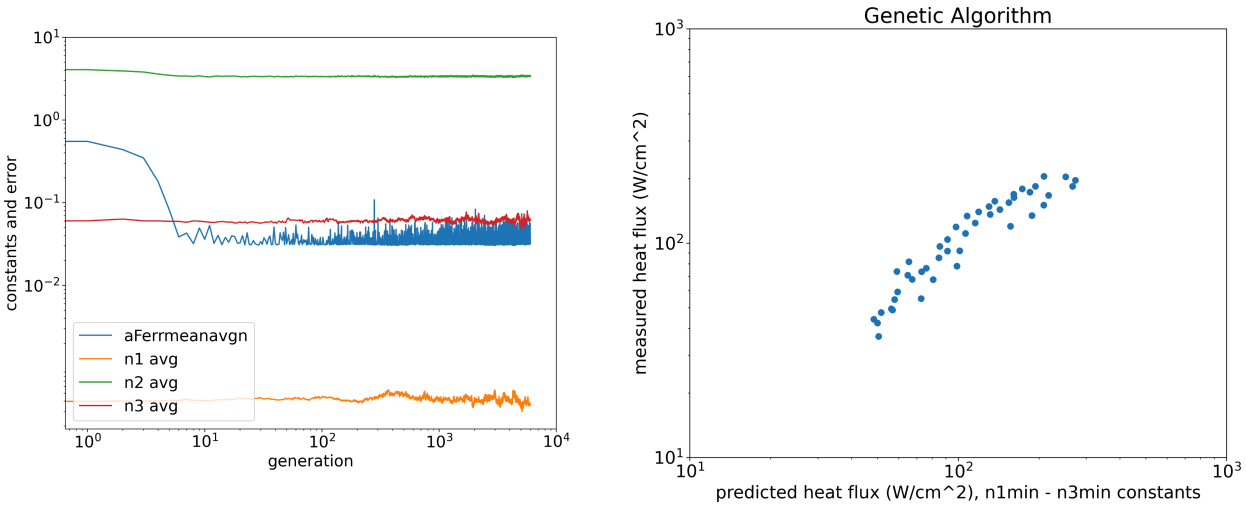


Figure 1.1. Convergence and performance of genetic algorithm for Group 1 heat-transfer data fitting.

Metric Type	n_1	n_2	n_3	aF err_mean
Ending	0.00037	3.3674	0.06214	0.03732
Minimum	0.00052	3.2958	0.05640	0.03029
Time Avg.	0.00041	3.3684	0.06125	0.03562

Table 2.1. Genetic algorithm optimization metrics and error statistics for Group 1.

RMS deviation	20.07%
Mean absolute error	14.86%
Maximum error	52.91%

Table 3.1. RMS, mean absolute, and maximum error values for Group 1 fit.

Group 2

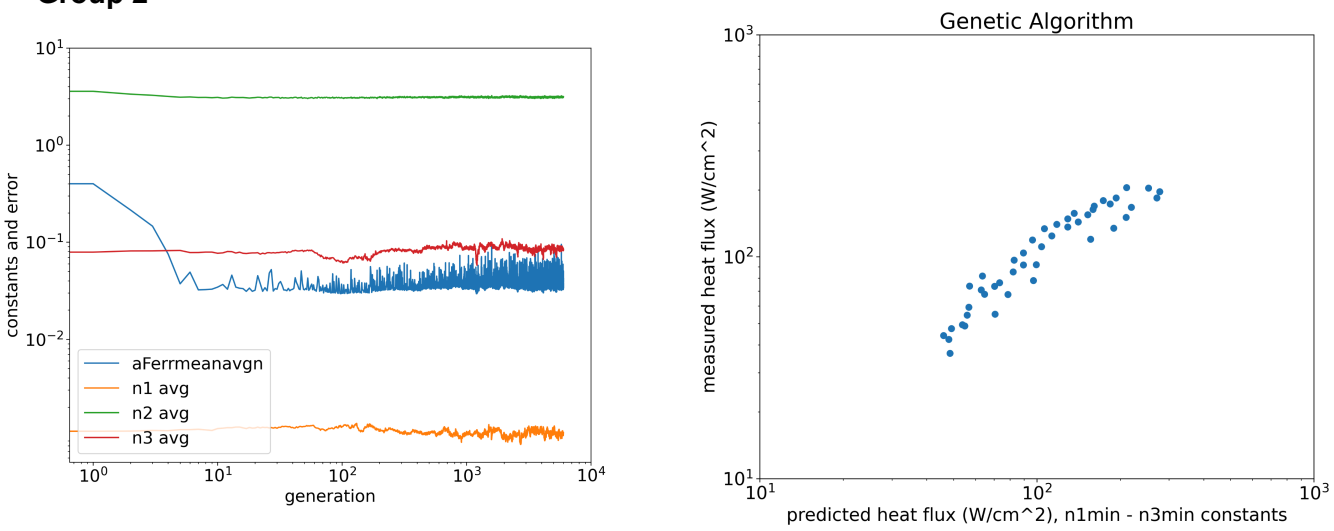


Figure 1.2. Convergence and performance of genetic algorithm for Group 2 heat-transfer data fitting.

Metric Type	n_1	n_2	n_3	aF err_mean
Ending	0.00109	3.1274	0.0828	0.0374
Minimum	0.00113	3.0902	0.0609	0.03029
Time Avg.	0.00107	3.1154	0.0865	0.0374

Table 2.2. Genetic algorithm optimization metrics and error statistics for Group 2.

RMS deviation	18.95%
Mean absolute error	14.37%
Maximum error	47.43%

Table 3.2. RMS, mean absolute, and maximum error values for Group 1 fit.

Group 3

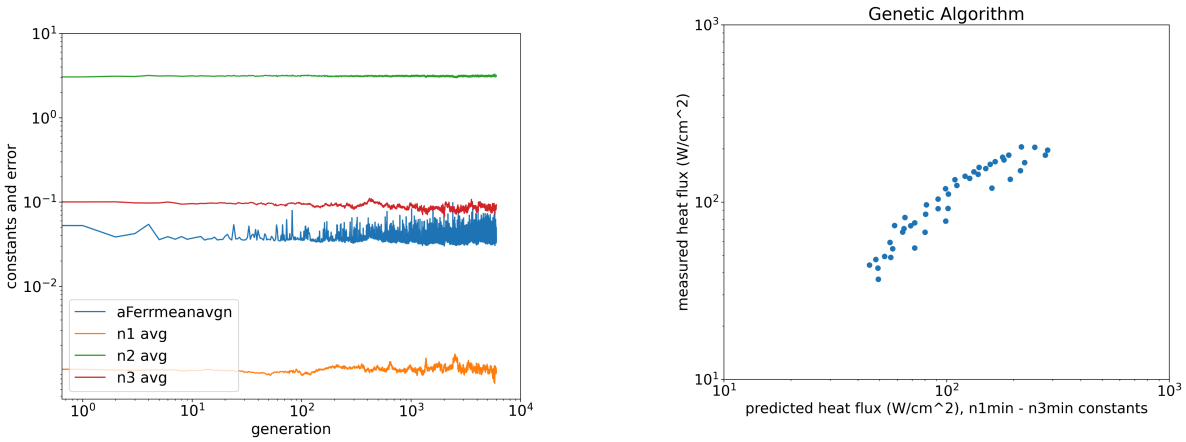


Figure 1.3. Convergence and performance of genetic algorithm for Group 3 heat-transfer data fitting.

Metric Type	n_1	n_2	n_3	aF err_mean
Ending	0.00098	3.1384	0.09131	0.03427
Minimum	0.00108	3.1027	0.06937	0.03016
Time Avg.	0.00106	3.1201	0.08567	0.03746

Table 2.3. Genetic algorithm optimization metrics and error statistics for Group 3

RMS deviation	19.91%
Mean absolute error	14.88%
Maximum error	50.86%

Table 3.3. RMS, mean absolute, and maximum error values for Group 3 fit.

Group 4

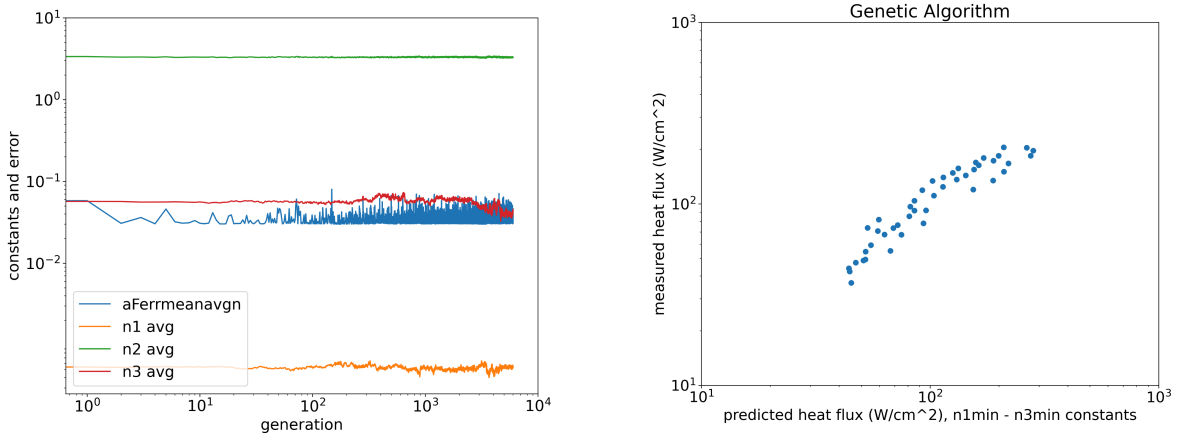


Figure 1.4. Convergence and performance of genetic algorithm for Group 4 heat-transfer data fitting.

Metric Type	n_1	n_2	n_3	aF err_mean
Ending	0.00054	3.2713	0.04433	0.03227
Minimum	0.00062	3.2466	0.05665	0.02986
Time Avg.	0.00052	3.3005	0.05043	0.03470

Table 2.3. Genetic algorithm optimization metrics and error statistics for Group 4

RMS deviation	19.29%
Mean absolute error	14.47%
Maximum error	49.69%

Table 3.4. RMS, mean absolute, and maximum error values for Group 4 fit.

Both Group 1 and Group 5 successfully achieved the target algorithm error of less than 0.035, converging to nearly identical optimized constants despite different initial guesses. This strong agreement indicates that the genetic algorithm consistently identified the same optimum for this correlation form. Among all cases, Group 5 produced the lowest algorithm error (0.0286), representing the best overall convergence and model performance.

A notable finding is that the root-mean-square (RMS) deviation remained approximately constant at 19–20%, irrespective of convergence quality Groups 1, 2, 3, and 5 all exhibited RMS values within this narrow range (19.0–20.1%), even though their algorithm errors varied substantially (0.0286–0.0694). The best-performing case, Group 5, achieved an algorithm error of 0.0286 but still yielded an RMS deviation of 19.29%. This persistent RMS value suggests that the limitation arises not from optimization inefficiency but from different fundamental factors.

Task 1.3

For this task, a five-parameter genetic algorithm optimization was performed to find the constants for the extended correlation. Running for 5,000 generations with a population size of 45, the algorithm converged to the optimal parameters as shown in **Table 2.4**. A surface plot heat flux versus gravity and surface tension at 10kPa shows decreased heat transfer at higher effective acceleration as shown in **Figure 1.6**.

Metric Type	Population-averaged at minimum error
n_1	0.000607
n_2	2.980
n_3	0.229
n_4	0.602
n_5	0.223

Table 2.4. Genetic algorithm optimization metrics for the five constant model.

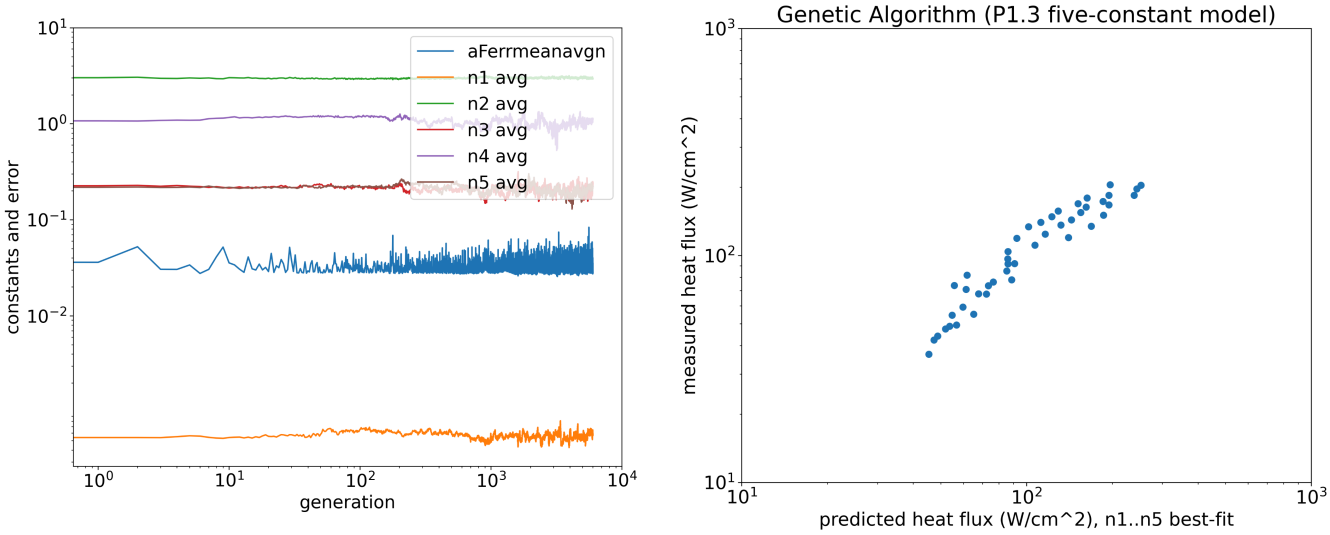


Figure 1.5. Convergence and performance of genetic algorithm for heat-transfer data fitting with five constants.

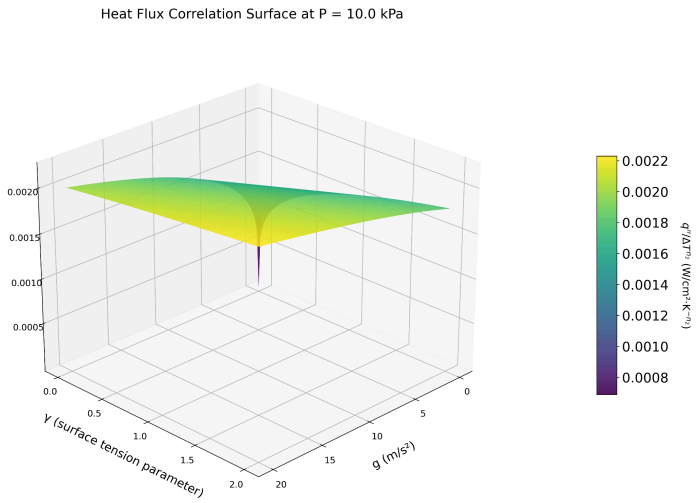


Figure 1.6. Surface plot of q"(T_w - T_{sat})^{n2} at pressure of 10kPa

Task 2.1

We modified P1.2 and wrote the python code for P2. We changed the error equation, initial values and split data into training(80%) and validation data(20%) and ran the model successfully.

Task 2.2

This task involved implementing a genetic algorithm to fit experimental heat transfer data.

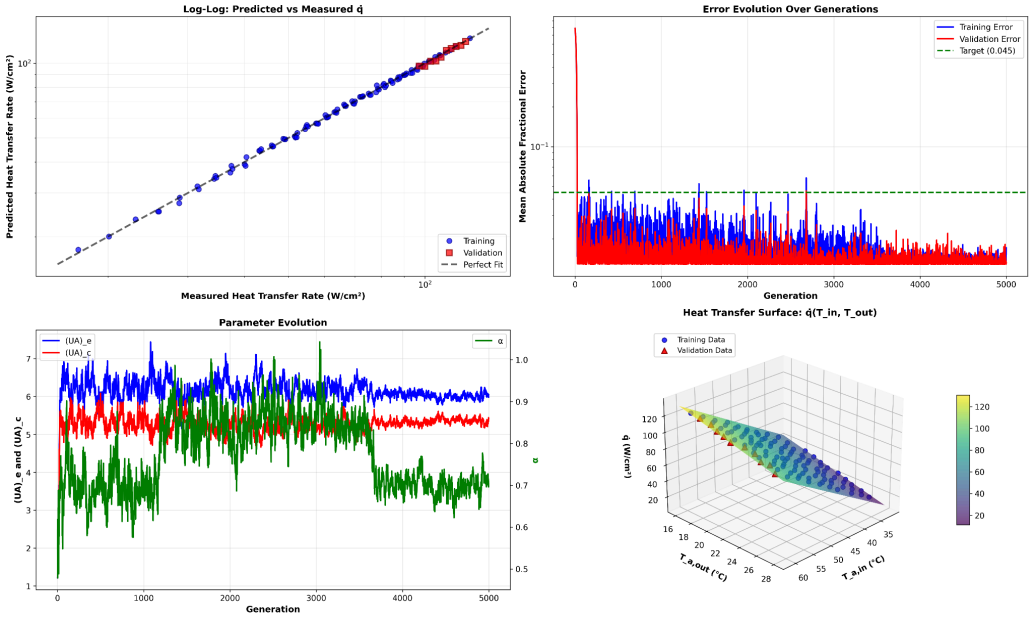


Figure 2.1. Model training and performance results.
Predicted vs. measured heat-transfer rates show strong agreement (top-left). Training and validation errors converge toward the target (top-right). Model parameters stabilize over generations (bottom-left). The heat-transfer surface fits training and validation data well (bottom-right) for UA(c) and UA(e) and sigma 1.0, 1.0, and 0.5 respectively.

The following correlation constants were used to compare between 4 different sets. To accomplish this task provided code was edited to include the following groups for the different n-values.

	n1(UA)c	n2 (UA)e	n3(sigma)
Group 1	1.0	1.0	0.5
Group 2	5.0	5.0	0.6
Group 3	10.0	10.0	0.4
Group 4	100	100	0.4

Table 2.1. Correlation constants for genetic-algorithm comparison between four data groups.

The attached plots represent results for each group showing the constants and error.

Group 1:

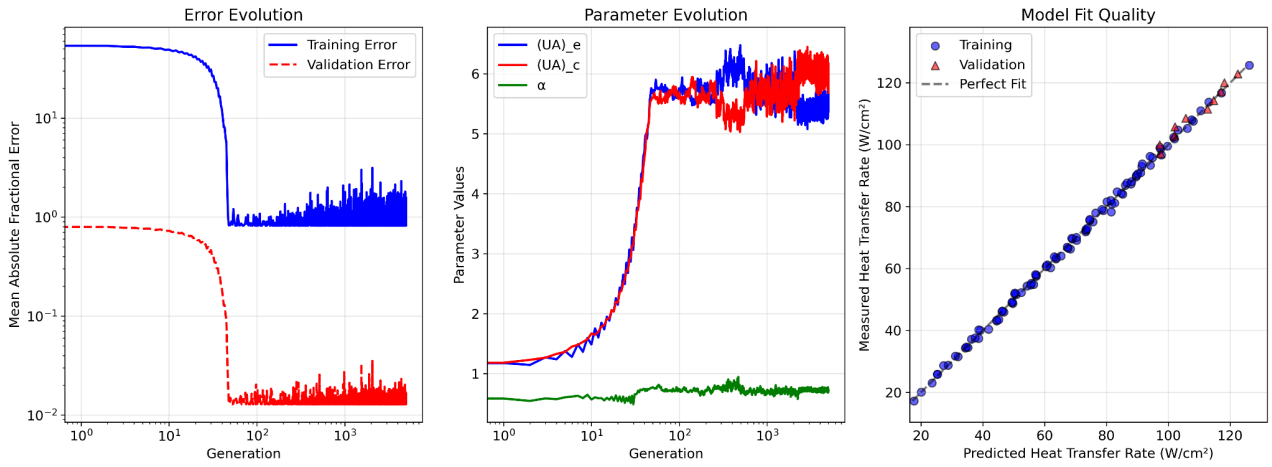


Figure 2.2. Convergence and performance of genetic algorithm for Group 1 heat-transfer data fitting.

	Training Data Error	Validation Data Error
RMS deviation	1.73 %	1.84%
Mean absolute error	1.35 % (<4.5%)	1.41% (<4.5 %)
Maximum error	4.79%	3.53%

Table 2.2. RMS, mean absolute, and maximum error values of both training and validation data for Group 1 fit.

Group 2:

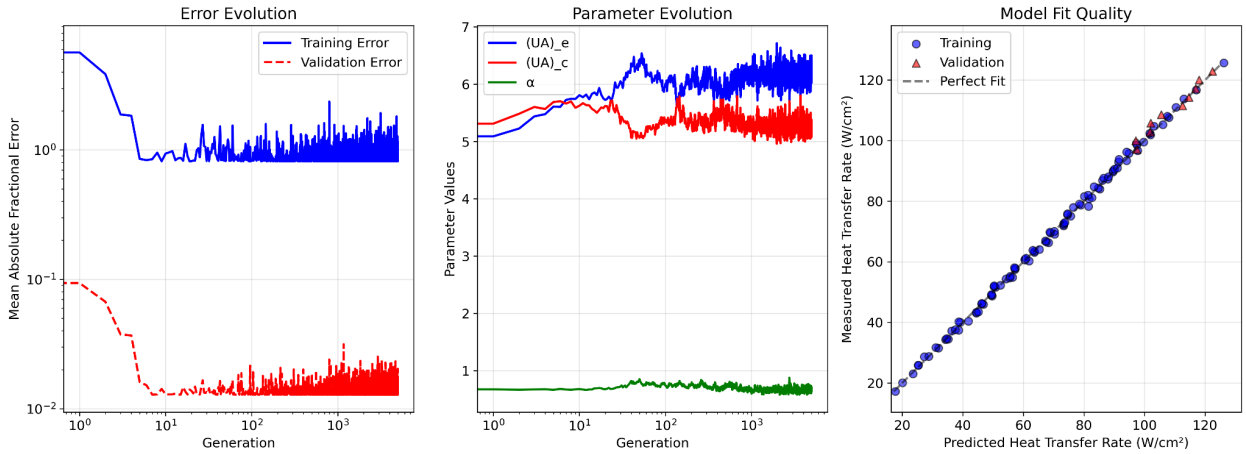


Figure 2.3. Convergence and performance of genetic algorithm for Group 2 heat-transfer data fitting.

	Training Data Error	Validation Data Error
RMS deviation	1.72 %	1.79%
Mean absolute error	1.35 % (<4.5%)	1.39% (<4.5 %)
Maximum error	5.0 %	3.45%

Table 2.3. RMS, mean absolute, and maximum error values of both training and validation data for Group 2 fit.

Group 3:

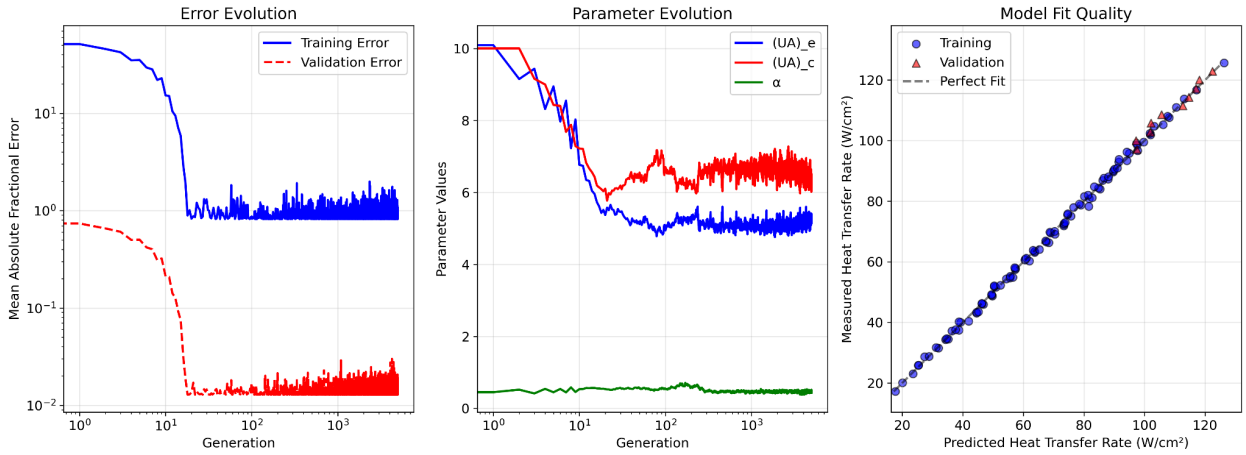


Figure 2.4. Convergence and performance of genetic algorithm for Group 3 heat-transfer data fitting.

	Training Data Error	Validation Data Error
RMS deviation	1.73 %	1.81%
Mean absolute error	1.35 % (<4.5%)	1.40 % (<4.5 %)
Maximum error	4.83 %	3.50%

Table 2.4. RMS, mean absolute, and maximum error values of both training and validation data for Group 3 fit.

Group 4:

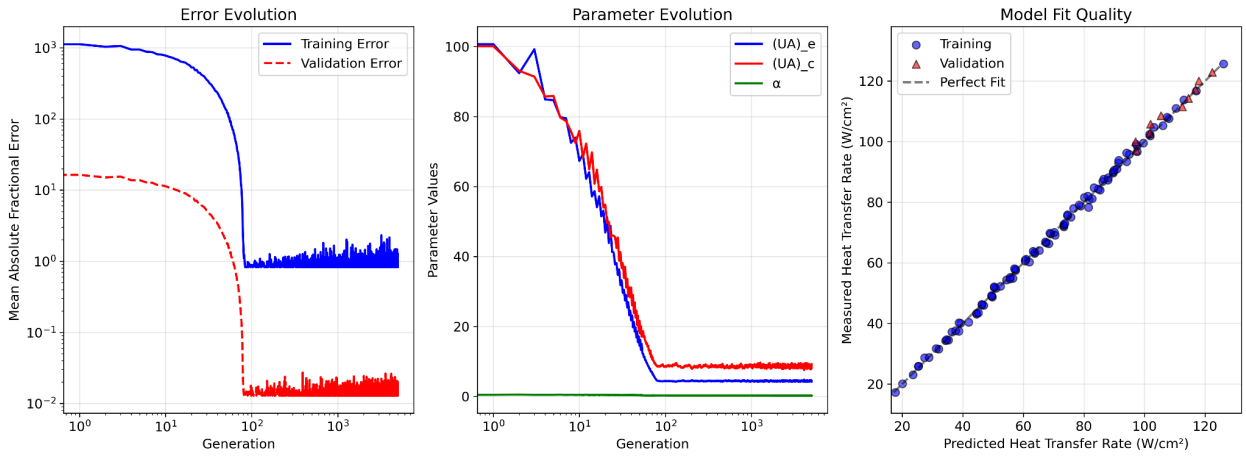


Figure 2.5. Convergence and performance of genetic algorithm for Group 4 heat-transfer data fitting.

	Training Data Error	Validation Data Error
RMS deviation	1.73 %	1.86%
Mean absolute error	1.35 % (<4.5%)	1.42 % (<4.5 %)
Maximum error	4.78 %	3.56%

Table 2.5. RMS, mean absolute, and maximum error values of both training and validation data for Group 4 fit.

In summary, across all groups, the genetic algorithm consistently achieved an RMS deviation of approximately 1.7%, a mean absolute error of about 1.35%, and a maximum error below 5%, successfully meeting the target normalized error criterion Ferr<0.045 (4.5%). The mean n1-n2-values converged to around 5 for all cases and 0.5 for n3, demonstrating the algorithm's insensitivity to initial guesses over four orders of magnitude (from 1 to 100). Group 2 produced the most stable and lowest overall error, while Group 4 confirmed that even extreme initializations ultimately converged to a similar optimum. Increasing the number of generations (NGEN) improved convergence smoothness and reduced error oscillations. The log–log plots of predicted versus measured heat-transfer rates showed near-linear behavior (slope ≈ 1), and the surface plot of q˙, T_s,in , T_s, out closely matched experimental trends, confirming the robustness and accuracy of the GA-based fitting.

Task 2.3

i) The work was organized into two phases. Initially, we jointly analyzed the provided codebase and set up a GitHub repository with separate branches for independent contributions. Subsequently, we collaboratively drafted the write-ups, meeting to align on structure and finalize formatting.

ii) We modified P1.2 into P2.2 by switching to a new three-column dataset [Tin,Tout,q˙] introducing an 80/20 train–validation split, and replacing the earlier power-law model with a heat-pipe energy model parameterized by (UA)e, (UA)c, and α, including a radiative term hr=σ(Tin,K+Tout,K), where σ(Tin,K+Tout,K)(Tin,K2+Tout,K2) computed in Kelvin based on the project's Eq. 2.6. The genetic algorithm now evolves only three genes with positivity clamps and a safeguarded "no-survivor" path, with slightly tuned mutation (~9%). Finally, we expanded diagnostics: per-generation parameter means, training and validation error curves, a predicted-vs-measured scatter for both splits, MAE summaries.

Appendix